

Fraud Detection Platform

Executive Rollup Report

Prepared by Nandagopal · Generated 2026-09-13 17:38 UTC · Auto-populated from NB1-NB9's real on-disk outputs -- zero-fabrication policy

Champion model: **CatBoost** | Model tier: **Tier 2 -- Moderate materiality -- proportionate, standard oversight**
| Real dataset: **284,807 transactions, 492 confirmed frauds**

Executive Summary

Metric	Real Value	Source
Champion CV PR-AUC (5-fold)	0.8482	NB1
Temporal-split PR-AUC (honest)	0.7692	NB1
Cost-optimal threshold	0.0443	NB1
Real precision / recall (OOF)	69.51% / 82.93%	NB1
Savings vs. no model	EUR 199,995.35 (approx. USD 231,834.61)	NB1
Savings vs. naive 0.5 threshold	EUR 7,865.08 (approx. USD 9,117.20)	NB1
Model tier (composite rubric)	Tier 2 (5/9)	NB6
BCBS 239 data-quality gate	6/6 checks passed, 284,807 rows	NB7
Governance sign-off readiness	21/30 checks Pass, 5 outside pipeline scope	NB9

SMART Decisions by Role (auto-derived from real flags across NB1-NB9)

CEO / Board -- Approve continued investment in the fraud-detection program

Real basis: Measured savings vs. no model: EUR 199,995.35 (approx. USD 231,834.61) over the real 284,807-transaction sampled window. Classified Moderate materiality -- proportionate, standard oversight (Tier 2, composite score 5/9).

Conditional -- 4 of 30 governance sign-off checks (NB9) are marked outside this pipeline's scope -- real organizational facts, not model gaps. (Source: NB1, NB6, NB9)

CFO -- Book projected fraud-loss savings into the budget

Real basis: Real total cost by scenario -- no model EUR 265,164.35 (approx. USD 307,378.51), naive threshold EUR 73,034.07 (approx. USD 84,661.10), cost-optimal EUR 65,169.00 (approx. USD 75,543.90).

Conditional -- Figures are real totals over the sampled ~48-hour window, not annualized -- NB6 explicitly disclosed that annualizing would require a representativeness assumption this project has not sourced. (Source: NB1, NB6)

Chief Risk Officer (CRO) -- Accept the model into risk appetite at Tier 2

Real basis: Worst-case NB5 stress-grid loss is 21.3x today's real cost (EUR 65,169.00 (approx. USD 75,543.90) -> EUR 1,390,976.41 (approx. USD 1,612,419.85)). 2 open regulatory-scrutiny flags.

Conditional -- NB2 real drift monitoring any_alert=True (worst real feature PSI 1.4298) -- resolve before treating Tier 2 as steady-state. (Source: NB2, NB5, NB6)

Chief Compliance Officer (CCO) -- Sign off on the regulatory applicability assessment

Real basis: 4 real frameworks mapped (BSA/SAR, Reg E, GLBA Safeguards, PSD2 SCA). BCBS 239 data-quality gate: 6/6 checks passed on 284,807 real rows.

Conditional -- Data architecture, Completeness remain limitation-disclosed, not evidenced. (Source: NB7, NB8)

CTO / Head of Data & Analytics -- Approve production deployment of the real-time scoring service

Real basis: Real local API: p99 client latency 5.10 ms. Training/serving consistency check: max abs diff 0.0 over 500 real rows.

Go -- Health check and train/serve consistency both real-verified PASS; latency is a local signal only, not yet a reachable production deployment. (Source: NB4)

Head of Fraud Operations -- Adopt the cost-optimal decision threshold operationally

Real basis: Threshold 0.0443 -> real precision 69.51%, real recall 82.93%. Human-review band (ASSUMPTION +/-0.02): 51 of 284,807 real transactions routed for manual review.

Go -- Threshold is real and vectorized-search-verified (NB1); review band is a disclosed ASSUMPTION, not a measured optimum. (Source: NB1, NB8)

Model Risk Manager -- Complete Tier 2 (second-line) model-risk documentation

Real basis: NB9 real evidence: Tier 2 checks 7/7 Pass (methodology, environment pinning, RANDOM_SEED=42 consistency, fairness-scope disclosure, known limitations, challenger model, model card).

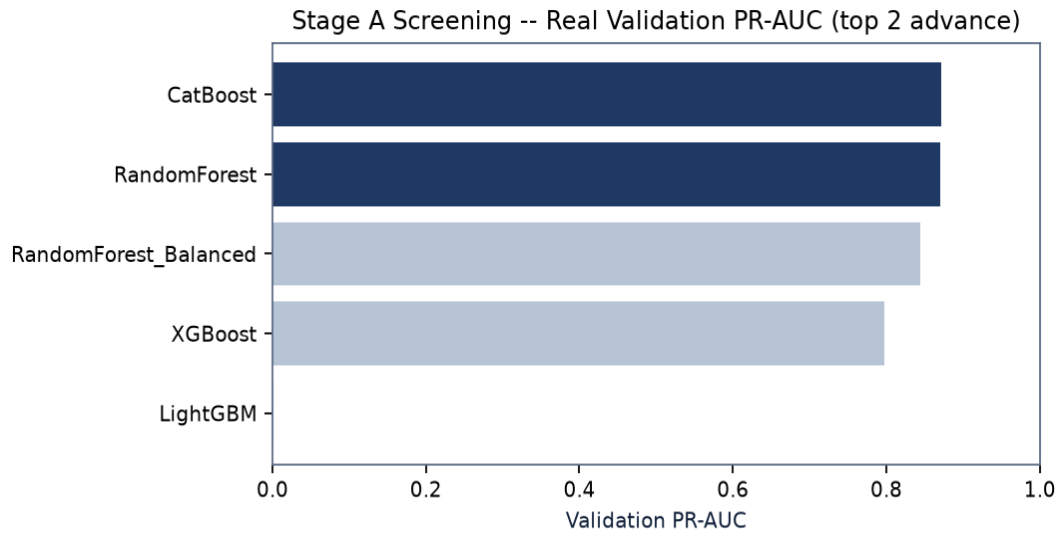
Go -- All 7 Tier 2 checks pass on real evidence (NB9) -- ready for a real human Model Risk Manager signature. (Source: NB9)

Technical Lead / Data Science -- Resolve the open external-benchmark investigate flag

Real basis: investigate_flag=True, precision_gap=-23.82% vs. external reference (your precision 69.51% vs. reference 93.33%).

Conditional -- Large positive gaps may indicate leakage; large negative gaps may indicate a bug — investigate before trusting either. (Source: NB1)

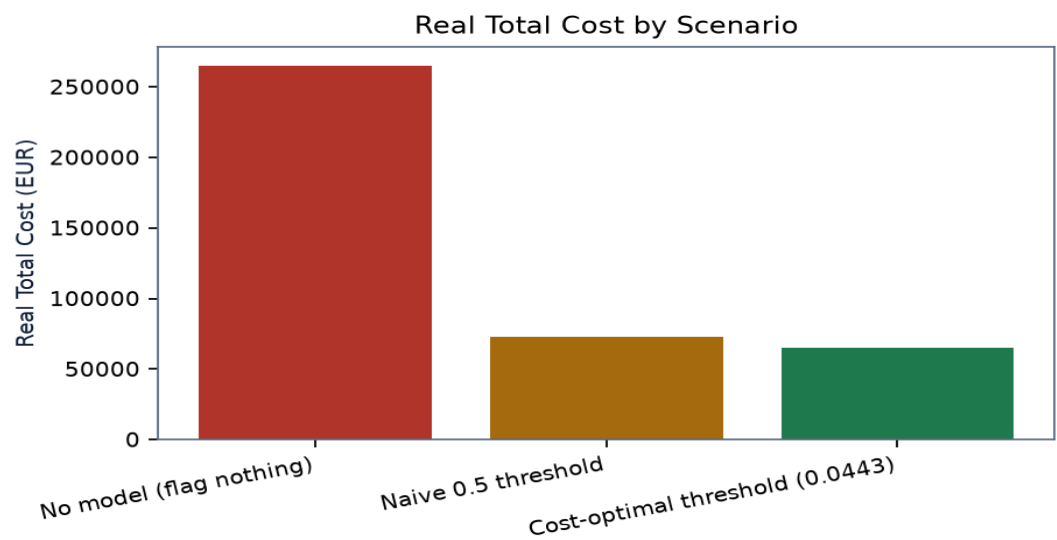
01-02. Model Screening & Champion Validation (NB1)



Champion: **CatBoost** (mean CV PR-AUC 0.8482) vs. runner-up **RandomForest** (0.8448). Honest temporal-split PR-AUC is 0.7692 -- a real, disclosed divergence from the shuffled CV number, since the temporal split trains on the earlier portion and tests on strictly later transactions.

External benchmark check: your precision 69.51% vs. reference 93.33% (gap -23.82%); your recall 82.93% vs. reference 74.67% (gap 8.26%). `investigate_flag=True`. Large positive gaps may indicate leakage; large negative gaps may indicate a bug — investigate before trusting either.

03. Cost-Optimal Threshold & Financial Impact (NB1)



Scenario	Real Total Cost
No model (flag nothing)	EUR 265,164.35 (approx. USD 307,378.51)
Naive 0.5 threshold	EUR 73,034.07 (approx. USD 84,661.10)
Cost-optimal threshold (0.0443)	EUR 65,169.00 (approx. USD 75,543.90)

Cost model (Master Playbook Section 10, real sourced constants): false-negative cost = amount lost x 4.41 (LexisNexis True Cost of Fraud multiplier); false-positive cost = flagged amount x 9.2% (Aite-Novarica/Statista via Riskified). EUR/USD 1.1592 (ECB euro foreign exchange reference rate, 2026-09-11 fixing).

04-05. Governance Gates, Drift & Adversarial Robustness (NB2)

Gate 1 structural checks: 5/5 passed (all_passed=True). Gate 2 CV stability: True. Drift monitoring any_alert=**True** (worst real feature PSI 1.4298; score KS statistic 0.2022).

Adversarial boundary search: 3 of 492 real fraud cases evadable within a 90% amount-change budget (evasion rate 0.61%, median change for evasion 58%). Evaluated in-sample -- realistic held-out recall ceiling is 82.93%, not the in-sample 100% baseline.

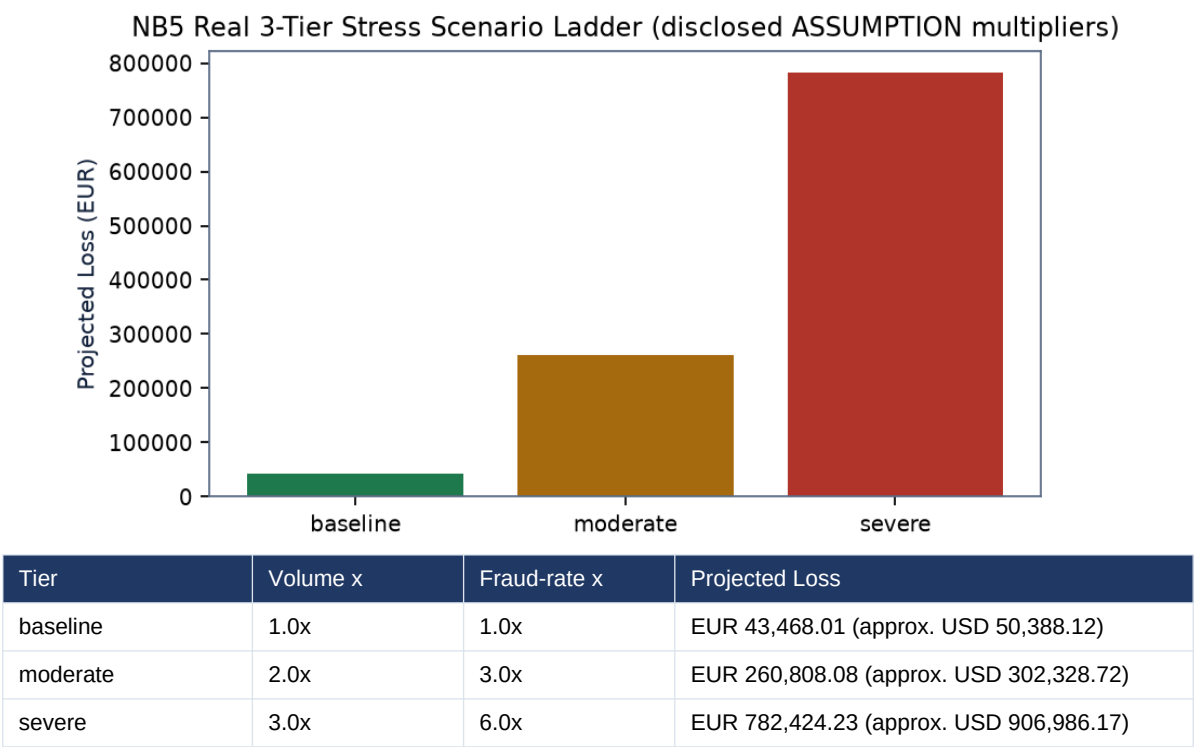
06. Deployment & Serving Consistency (NB4)

Real local API latency (client round-trip): mean 3.15 ms, p50 3.02 ms, p95 4.19 ms, p99 5.10 ms.

Training/serving consistency: max abs diff 0.0 over 500 real rows (passed=True). In-process FastAPI TestClient

-- real app/serialization overhead, NOT real network transport time.

07. Deepened Stress Testing (NB5)



Worst of 105 real swept combinations: volume x4.0, fraud-rate x8.0 -> projected loss EUR 1,390,976.41 (approx. USD 1,612,419.85).

08. Model Tiering Matrix (NB6)

Dimension	Real Input	Score
Financial Exposure	21.3x worst-case vs. current cost	2/3
Regulatory Scrutiny	2 real open flags	2/3
Customer Impact	179 real FPs / 284,807	1/3

Composite score 5/9 -> **Tier 2 (Moderate materiality -- proportionate, standard oversight)**. Retroactive finding: NB3's drift-monitoring history used an unjustified default tier=1; recommend tier=2 going forward.

09. BCBS 239 Data Governance (NB7)

Principle Group	Status	Evidence
Governance	evidenced	NB6 real Tier 2 (Moderate materiality -- proportionate, standard oversight); oversight cadence follows NB6's real recommended-next-steps for that tier.
Data architecture	limitation disclosed	Single anonymized source table (V1-V28 PCA components, Amount, Time, Class) -- no cross-source identifiers or taxonomy exist to integrate at this project's scope. Not fabricated.
Accuracy & integrity	evidenced	NB2 Gate 1 structural checks (real, all_passed=True); this notebook's own real, rerunnable data-quality gate (Section A, all_checks_passed=True).
Completeness	limitation disclosed	No business-line/legal-entity/region dimensions exist in this dataset (see Section B). Amount-band x hour-of-day (NB2's real concentration report) is the disclosed substitute.
Timeliness	evidenced	NB3's real append-only monitoring cadence + this notebook's own real append-only data-quality gate -- both designed to be rerun against each new real data batch, aggregation frequency scaling with how often new data arrives.
Adaptability	evidenced	NB6's real retroactive finding (recommending NB3's monitoring tier parameter be updated from its default) is a live, real example of the system adapting to a new governance decision.

10. Regulatory Applicability & Human Oversight (NB8)

Framework	Jurisdiction
Bank Secrecy Act (BSA) / Suspicious Activity Report (SAR) filing	US
Regulation E (Electronic Fund Transfer Act)	US
Gramm-Leach-Bliley Act (GLBA) Safeguards Rule	US
PSD2 Strong Customer Authentication & fraud-monitoring obligations	EU

Human-review band (ASSUMPTION +/-0.02 around the real threshold): 0.0243-0.0643. 51 of 284,807 real transactions fall in-band (0 fraud, 51 legit).

11. Governance Sign-Off Readiness (NB9)

Tier	Pass	Conditional	Outside Scope (TBD)	Total
1 -- Technical Lead	10	2	0	12
2 -- Model Risk Manager	7	0	0	7
3 -- CCO	2	1	2	5
4 -- Business Owner	2	1	3	6

Every Approve/Conditional/Reject decision and every Name/Date/Signature field in the underlying governance template is intentionally left blank -- this rollup assembles real evidence, it does not fabricate a human decision.